Fourier Transform of the Sine and Cosine Functions

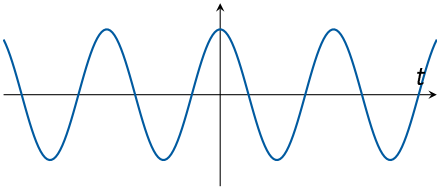
with an application to the driven oscillator

Sebastián

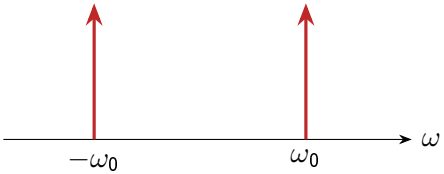
Physics Program
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Motivation



$f(t) = \cos \omega_0 t$



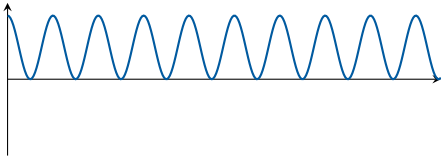
$F(\omega)$

A tuning fork, a string or an ideal spring oscillates at a single frequency. The Fourier transform is the tool that tells us which frequencies a signal contains. For the sine and the cosine we expect "just one", and we will see that the mathematical answer is two Dirac deltas, at $+\omega_0$ and at $-\omega_0$.

The idea behind the transform

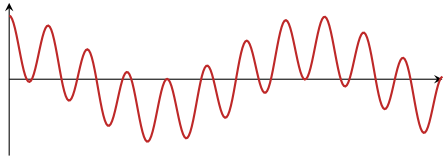
To find out whether a signal $f(t)$ contains the frequency ω, we multiply it by an oscillation of that frequency and integrate.

$\cos(2t) \cdot \cos(2t)$



Same frequency: the product is always positive and the integral grows.

$\cos(2t) \cdot \cos(2.6t)$



Different frequencies: the product changes sign and the integral cancels on average.

So that we don't have to test with sine and cosine separately, we use the complex exponential $\mathrm{e}^{-\mathrm{i}\omega t} = \cos\omega t - \mathrm{i}\sin\omega t$, which contains both.

Definition

With that idea we define

$$F(\omega) = \int_{-\infty}^{\infty} f(t)\, \mathrm{e}^{-\mathrm{i}\omega t}\, \mathrm{d}t.$$

In general $F(\omega)$ is a complex number: its modulus tells us how much of the frequency ω there is, and its argument tells us with what phase it appears.

The signal is recovered with the inverse transform,

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega)\, \mathrm{e}^{\mathrm{i}\omega t}\, \mathrm{d}\omega,$$

which reads as "f is a sum (integral) of oscillations $\mathrm{e}^{\mathrm{i}\omega t}$, each one with weight $F(\omega)/2\pi$".

Other books put the factor 2π somewhere else, or use the frequency f in Hz with $\mathrm{e}^{-2\pi \mathrm{i} f t}$. The constants change, not the form of the results.

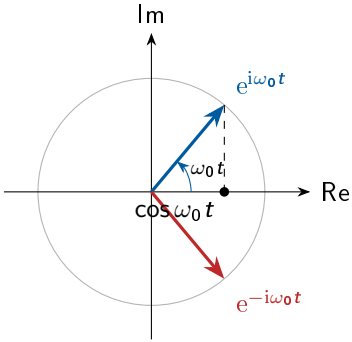
The cosine as a sum of two phasors

From Euler's formula, $e^{\pm i\theta} = \cos\theta \pm i \sin\theta$, we get

$$\cos\omega_0 t = \frac{e^{i\omega_0 t} + e^{-i\omega_0 t}}{2}, \qquad \sin\omega_0 t = \frac{e^{i\omega_0 t} - e^{-i\omega_0 t}}{2i}.$$

$e^{i\omega_0 t}$ is a vector of length 1 that rotates counterclockwise with angular velocity ω_0, and $e^{-i\omega_0 t}$ rotates clockwise.

When we add them, the imaginary parts cancel and what is left is a real number oscillating along the horizontal axis: the cosine. That is why both frequencies, $+\omega_0$ and $-\omega_0$, will appear in the spectrum.



The problem: the integral does not converge

If we try to compute the transform directly:

$$\int_{-\infty}^{\infty} \cos(\omega_0 t)\, \mathrm{e}^{-\mathrm{i}\omega t}\, \mathrm{d}t = \frac{1}{2}\int_{-\infty}^{\infty} \mathrm{e}^{-\mathrm{i}(\omega-\omega_0)t}\, \mathrm{d}t + \frac{1}{2}\int_{-\infty}^{\infty} \mathrm{e}^{-\mathrm{i}(\omega+\omega_0)t}\, \mathrm{d}t$$

Both integrals are of the same type, $\int_{-\infty}^{\infty} \mathrm{e}^{-\mathrm{i}\Omega t}\, \mathrm{d}t$:

- if $\Omega \neq 0$, the integrand keeps rotating and does not decay, so the integral oscillates and has no limit;
- if $\Omega = 0$, the integrand equals 1 everywhere and the integral is infinite.

The root of the problem is that the cosine lasts forever and does not decay: $\int |\cos \omega_0 t|\, \mathrm{d}t = \infty$. The usual transform is only guaranteed to exist for integrable functions.

To get around this, we observe the signal for a finite time, compute the transform (which does exist), and then let that time grow.

Finite window: the calculation

We integrate only between $-T$ and T:

$$\int_{-T}^{T} \mathrm{e}^{-\mathrm{i}\Omega t}\,\mathrm{d}t = \left[\frac{\mathrm{e}^{-\mathrm{i}\Omega t}}{-\mathrm{i}\Omega}\right]_{-T}^{T} = \frac{\mathrm{e}^{-\mathrm{i}\Omega T} - \mathrm{e}^{\mathrm{i}\Omega T}}{-\mathrm{i}\Omega}$$
$$= \frac{-2\mathrm{i}\sin(\Omega T)}{-\mathrm{i}\Omega} = \frac{2\sin(\Omega T)}{\Omega}.$$

In the second line we used $\mathrm{e}^{\mathrm{i}a} - \mathrm{e}^{-\mathrm{i}a} = 2\mathrm{i}\sin a$.

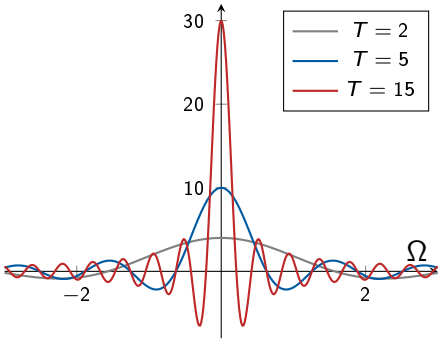
We call $D_T(\Omega) = \dfrac{2\sin(\Omega T)}{\Omega}$. The result is now an ordinary function; what remains is to see what happens to it as $T \to \infty$.

Finite window: what happens to D_T

- **Height.** Near $\Omega = 0$, $\sin(\Omega T) \approx \Omega T$, so $D_T(0) = 2T$. It grows without bound.
- **Width.** The first zero is where $\Omega T = \pi$, that is, $\Omega = \pi/T$. The peak gets narrower.
- **Area.** With $u = \Omega T$,

$$\int_{-\infty}^{\infty} \frac{2\sin \Omega T}{\Omega}\, \mathrm{d}\Omega = 2\int_{-\infty}^{\infty} \frac{\sin u}{u}\, \mathrm{d}u = 2\pi,$$

which does not depend on T.



A peak that gets taller and narrower, but with a fixed area 2π: that is exactly what defines a Dirac delta.

The Dirac delta

$\delta(\Omega)$ is not a function in the usual sense. It is defined by what it does inside an integral:

$$\int_{-\infty}^{\infty} \delta(\Omega)\,\varphi(\Omega)\,\mathrm{d}\Omega = \varphi(0) \qquad \text{for every smooth function } \varphi.$$

Properties we will use:

$$\int_{-\infty}^{\infty} \delta(\Omega)\,\mathrm{d}\Omega = 1, \qquad g(\Omega)\,\delta(\Omega - a) = g(a)\,\delta(\Omega - a).$$

Back to the window: as $T \to \infty$, D_T concentrates at $\Omega = 0$ with area 2π, so that

$$\int_{-\infty}^{\infty} D_T(\Omega)\,\varphi(\Omega)\,\mathrm{d}\Omega \longrightarrow 2\pi\,\varphi(0) \qquad \Longrightarrow \qquad \boxed{\int_{-\infty}^{\infty} \mathrm{e}^{-\mathrm{i}\Omega t}\,\mathrm{d}t = 2\pi\,\delta(\Omega)}$$

Transform of the cosine

We start from

$$\cos\omega_0 t = \tfrac{1}{2}\,\mathrm{e}^{\mathrm{i}\omega_0 t} + \tfrac{1}{2}\,\mathrm{e}^{-\mathrm{i}\omega_0 t}.$$

The transform of each exponential follows from the previous result with $\Omega = \omega \mp \omega_0$:

$$\mathcal{F}\{\mathrm{e}^{\mathrm{i}\omega_0 t}\} = \int_{-\infty}^{\infty} \mathrm{e}^{-\mathrm{i}(\omega-\omega_0)t}\,\mathrm{d}t = 2\pi\,\delta(\omega-\omega_0), \qquad \mathcal{F}\{\mathrm{e}^{-\mathrm{i}\omega_0 t}\} = 2\pi\,\delta(\omega+\omega_0).$$

Since the transform is linear,

$$\boxed{\mathcal{F}\{\cos\omega_0 t\} = \pi\left[\delta(\omega-\omega_0) + \delta(\omega+\omega_0)\right]}$$

To check it, we apply the inverse transform and use the property of the delta:

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} \pi\left[\delta(\omega-\omega_0) + \delta(\omega+\omega_0)\right]\mathrm{e}^{\mathrm{i}\omega t}\,\mathrm{d}\omega = \frac{\mathrm{e}^{\mathrm{i}\omega_0 t} + \mathrm{e}^{-\mathrm{i}\omega_0 t}}{2} = \cos\omega_0 t.$$

Transform of the sine

The procedure is the same:

$$\sin\omega_0 t = \frac{1}{2\mathrm{i}}\,\mathrm{e}^{\mathrm{i}\omega_0 t} - \frac{1}{2\mathrm{i}}\,\mathrm{e}^{-\mathrm{i}\omega_0 t} \quad\Longrightarrow\quad \mathcal{F}\{\sin\omega_0 t\} = \frac{2\pi}{2\mathrm{i}}\left[\delta(\omega-\omega_0) - \delta(\omega+\omega_0)\right].$$

Multiplying numerator and denominator by i gives $\dfrac{1}{\mathrm{i}} = -\mathrm{i}$, so

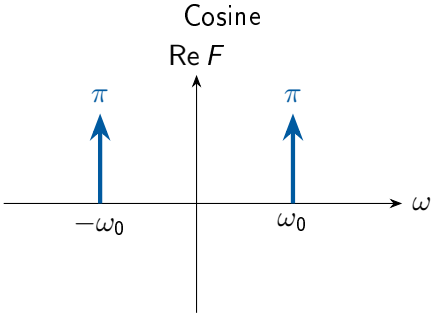
$$\boxed{\mathcal{F}\{\sin\omega_0 t\} = \mathrm{i}\pi\left[\delta(\omega+\omega_0) - \delta(\omega-\omega_0)\right]}$$

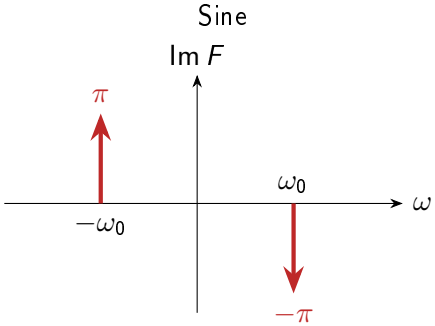
Check:

$$\frac{1}{2\pi}\,(\mathrm{i}\pi)\left(\mathrm{e}^{-\mathrm{i}\omega_0 t} - \mathrm{e}^{\mathrm{i}\omega_0 t}\right) = \frac{\mathrm{i}}{2}\,(-2\mathrm{i}\sin\omega_0 t) = \sin\omega_0 t.$$

Unlike the cosine, the result is purely imaginary.

Comparing the two spectra





- Same frequencies and same $|F(\omega)|$; only the phase changes.
- Even cosine $\Rightarrow$ real and even transform. Odd sine $\Rightarrow$ imaginary and odd transform.
- The height of each arrow is the weight (area) of the delta.

Interpretation

Negative frequencies. They are not a different physical thing: they appear because we write a real signal with complex exponentials, and both directions of rotation are needed for the sum to be real. Every real signal satisfies $F(-\omega) = \overline{F(\omega)}$, so the negative part carries no new information.

The phase. The sine is a delayed cosine: $\sin \omega_0 t = \cos\big(\omega_0 (t - t_0)\big)$ with $t_0 = \pi/(2\omega_0)$. A delay t_0 multiplies the transform by $\mathrm{e}^{-\mathrm{i}\omega t_0}$, which at $\omega = \pm\omega_0$ equals $\mathrm{e}^{\mp \mathrm{i}\pi/2} = \mp\mathrm{i}$. That explains the factors of i in the sine: the frequencies do not change, only the phase.

Why deltas? An oscillation that never ends has infinite total energy, but finite average power, $\langle A^2 \cos^2 \omega_0 t \rangle = A^2/2$. All that power sits at a single frequency, and a finite amount concentrated at one point is exactly a delta.

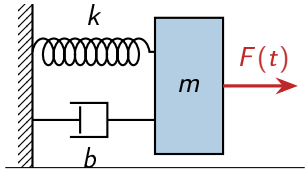
Application: the driven oscillator

A block attached to a spring, with friction, is pushed by a periodic force:

$$m\,\ddot{x} + b\,\dot{x} + k\,x = F_0\cos(\Omega t).$$

With what amplitude and what phase does the block oscillate?

Values: $m = 0.5\,\text{kg}$, $k = 20\,\text{N/m}$, $b = 0.5\,\text{kg/s}$, $F_0 = 1\,\text{N}$.



Transforming the equation

Differentiating $\mathrm{e}^{\mathrm{i}\omega t}$ is the same as multiplying it by $\mathrm{i}\omega$, so

$$\mathcal{F}\{\dot{x}\} = \mathrm{i}\omega X(\omega), \qquad \mathcal{F}\{\ddot{x}\} = -\omega^2 X(\omega).$$

We apply $\mathcal{F}$ to the equation and use the transform of the cosine on the right-hand side:

$$\left(k - m\omega^2 + \mathrm{i}b\omega\right) X(\omega) = \pi F_0 \left[\delta(\omega - \Omega) + \delta(\omega + \Omega)\right].$$

With $H(\omega) = k - m\omega^2 + \mathrm{i}b\omega$:

$$X(\omega) = \frac{\pi F_0}{H(\omega)} \left[\delta(\omega - \Omega) + \delta(\omega + \Omega)\right].$$

The differential equation has become an algebraic one.

Back to the time domain

In the inverse transform, the deltas do all the work:

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)\,\mathrm{e}^{\mathrm{i}\omega t}\,\mathrm{d}\omega = \frac{F_0}{2}\left[\frac{\mathrm{e}^{\mathrm{i}\Omega t}}{H(\Omega)} + \frac{\mathrm{e}^{-\mathrm{i}\Omega t}}{H(-\Omega)}\right] = \mathrm{Re}\left[\frac{F_0\,\mathrm{e}^{\mathrm{i}\Omega t}}{H(\Omega)}\right],$$

because the two terms are complex conjugates. Writing $H(\Omega) = |H|\mathrm{e}^{\mathrm{i}\varphi}$:

$$\boxed{x(t) = A\cos(\Omega t - \varphi)}, \qquad A = \frac{F_0}{\sqrt{(k - m\Omega^2)^2 + (b\Omega)^2}}, \qquad \tan\varphi = \frac{b\Omega}{k - m\Omega^2}.$$

The block oscillates at the frequency of the force. At resonance, $\Omega = \omega_0 = \sqrt{k/m}$, the amplitude is largest and the phase lag is $90°$. With a force $F_0\sin(\Omega t)$ we use the transform of the sine and get $A\sin(\Omega t - \varphi)$.

Checking with Python

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp

#System data
m, k, b, F0 = 0.5, 20.0, 0.5, 1.0
W = np.sqrt(k/m)            #driving frequency (rad/s)

#Fourier solution
A = F0/np.sqrt((k - m*W**2)**2 + (b*W)**2)
fi = np.arctan2(b*W, k - m*W**2)

#Numerical solution (starts at rest)
def equation(t, y):
    x, v = y
    return [v, (F0*np.cos(W*t) - b*v - k*x)/m]

t = np.linspace(0, 30, 3000)
sol = solve_ivp(equation, [0, 30], [0, 0],
                t_eval=t, max_step=0.01)
x = sol.y[0]

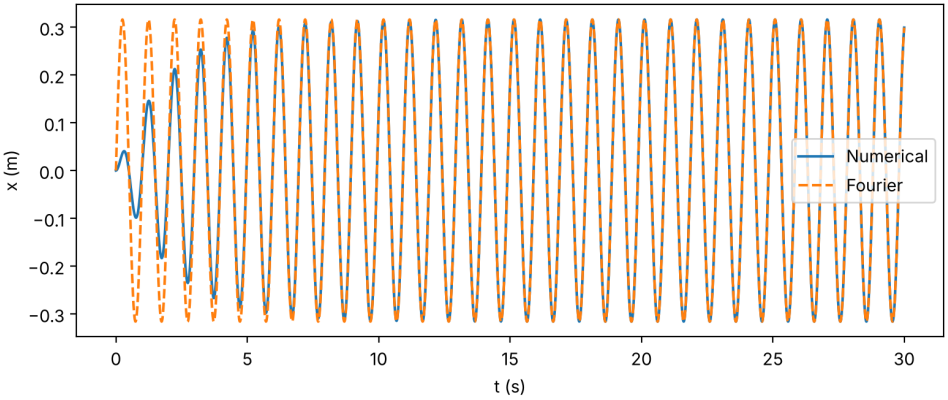
print("Fourier amplitude   =", round(A, 4), "m")
A_num = np.max(np.abs(x[t > 20]))      #after the transient
print("Numerical amplitude =", round(A_num, 4), "m")
print("Phase =", round(np.degrees(fi), 1), "degrees")
```

Fourier assumes the force has been acting forever. With Python we solve the full equation starting from rest, $x(0) = \dot{x}(0) = 0$, and compare.

Program output at resonance:

A Fourier	0.3162 m
A numerical	0.3162 m
φ	90.0°

Result



At first the numerical solution grows from zero: this is the transient, which depends on how the system starts and dies out because of friction. After about 10 s the two curves coincide, and the Fourier transform gives the steady motion exactly.

Conclusions

- The Fourier integral of the sine and the cosine does not converge in the usual sense; observing the signal for a finite time T and letting $T \to \infty$ gives two deltas at $\pm\omega_0$.
- The sine and the cosine have the same modulus. The cosine gives a real spectrum and the sine an imaginary one: only the phase changes.
- In the driven oscillator, the transform of the cosine turns the differential equation into an algebraic one, and the deltas solve the inverse transform. Python confirms the result once the transient has passed.

References

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- SciPy documentation, `scipy.integrate.solve_ivp`: `docs.scipy.org`.

Thank you